
AQUASENSE

Water Level Indicator

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INTRODUCTION

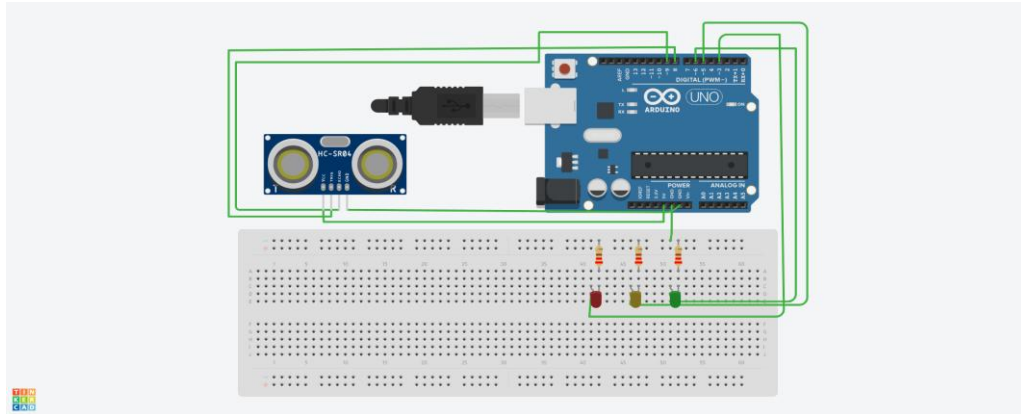
The **Smart Water Tank Level Monitoring System (Aqua Sense)** is designed and simulated using **Tinkercad Circuits** to monitor the water level inside a storage tank. The project aims to demonstrate the working of a smart water level monitoring system through virtual circuit simulation.

The system uses an **HC-SR04 ultrasonic sensor** to measure the distance between the sensor and the water surface. The measured distance is processed by an **Arduino Uno**, which determines the water level condition inside the tank. Based on the detected water level, the tank status is indicated using **LED indicators**, providing a the water level indication.

Components Required

Component	Quantity	Purpose
Arduino Uno	1	Main controller
HC-SR04 Ultrasonic Sensor	1	Measures water level distance
LEDs	3	Indicates tank status
Resistors	3	LED protection
Breadboard	1	Circuit connection

Circuit Diagram



Working

The HC-SR04 ultrasonic sensor continuously measures the distance between the sensor and the water surface. The Arduino processes the sensor readings and determines the water level condition. Based on the calculated level, LEDs are turned ON to indicate the tank status.

Green LED → high water level

Yellow LED → Medium water level

Red LED → Low water level

Algorithm

1. Start the System

The Arduino initializes the ultrasonic sensor and LED pins required for the operation.

2. Initialize Components

The HC-SR04 ultrasonic sensor is configured to send and receive ultrasonic signals, and the LEDs are set as output indicators.

3. Measure Distance

The ultrasonic sensor sends a sound pulse towards the water surface and receives the reflected echo. The Arduino calculates the distance between the sensor and the water surface using the time taken for the echo to return.

4. Determine Water Level Condition

The calculated distance is compared with predefined threshold values to identify the water level condition inside the tank.

5. Indicate Tank Status

Based on the detected water level, the corresponding LED is turned ON to indicate the tank status:

- High water level → Green LED ON
- Medium water level → Yellow LED ON
- Low water level → Red LED ON

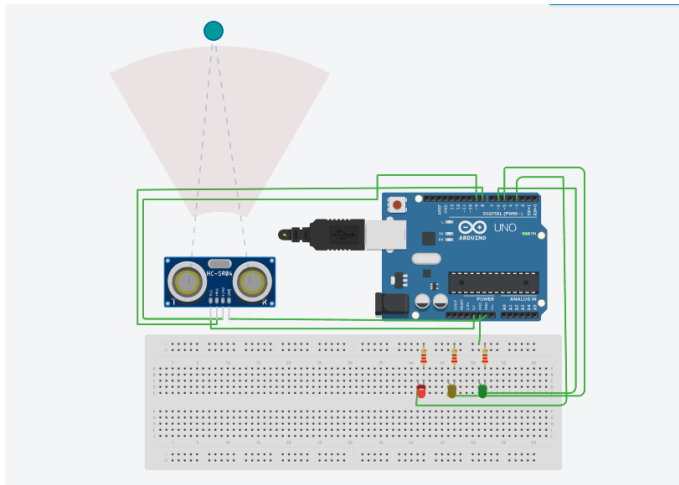
6. Repeat Monitoring Process

The system continuously repeats the measurement and indication process to provide real-time water level monitoring.

7. End

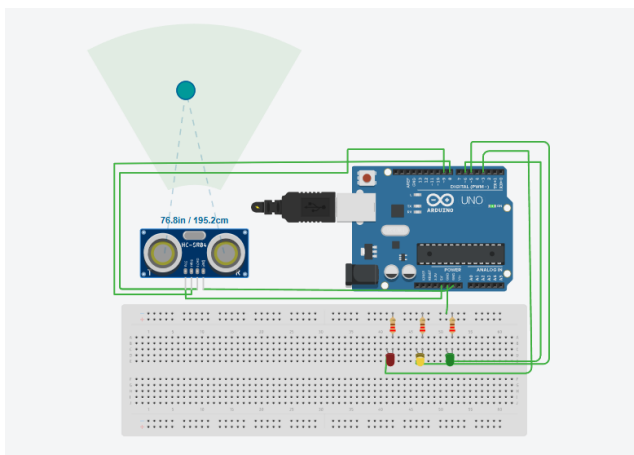
The process continues as long as the system remains powered on.

SIMULATION RESULTS



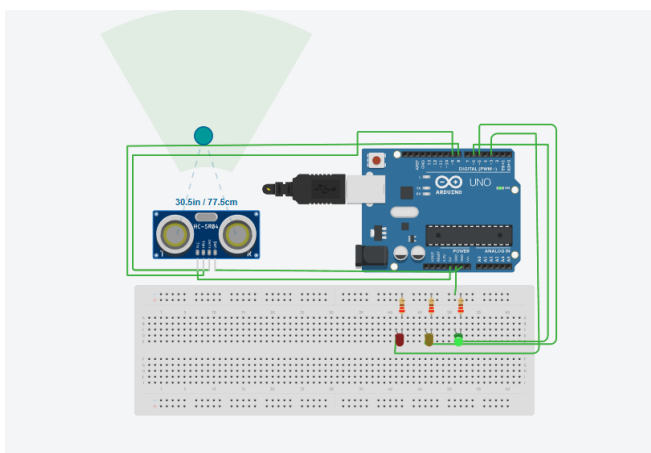
Case 1: Low Water Level Indication (Red LED Blinking)

In the first simulation case, the water level inside the tank is detected as **low** based on the distance measured by the **HC-SR04 ultrasonic sensor**. The Arduino processes the sensor data and identifies that the water level has reached below the predefined threshold value. As an indication of the low water level condition, the **red LED starts blinking**



Case 2: Medium Water Level Indication (Yellow LED Blinking)

In the second simulation case, the water level inside the tank is detected as **medium** based on the distance measured by the **HC-SR04 ultrasonic sensor**. The Arduino processes the sensor readings and determines that the water level is within the medium range based on the predefined threshold values.



Case 3: High Water Level Indication (Green LED Blinking)

In the third simulation case, the water level inside the tank is detected as **high** based on the distance measured by the **HC-SR04 ultrasonic sensor**. The Arduino processes the sensor readings and identifies that the water level has reached the high level range according to the predefined threshold values.

As an indication of the high-water level condition, the **green LED starts blinking**.

CONCLUSION

The Arduino-based water level indicator using the HC-SR04 ultrasonic sensor accurately monitors the water level and displays it using LEDs. It is a simple, reliable, and cost-effective system that helps prevent water overflow and improves efficient water management.